

**Project Title:** House Price Prediction

**Objective:** The goal of this project is to build a machine learning model that predicts house prices based on key features such as location, size, and amenities.

**Approach:**

- Collected and cleaned housing data.
- Performed exploratory data analysis (EDA) including correlation heatmaps.
- Trained regression models using scikit-learn.
- Evaluated performance with metrics like  $R^2$  and Mean Squared Error.
- Saved the trained model (house\_price\_model.pkl) for reuse.

**Key Insights:**

- Strong correlations were observed between square footage, number of rooms, and price.
- Feature importance analysis highlighted that location and size are the most influential predictors.
- The model demonstrates reliable accuracy for practical use cases.

**Deliverables:**

- Jupyter Notebook (HousePricePrediction.ipynb) with full workflow.
- Trained model file (house\_price\_model.pkl).
- Visualizations saved in screenshots/.
- Documentation (README.md) and dependency list (requirements.txt).

**Conclusion:** This project provides a reproducible framework for predicting house prices. It can be extended with more advanced models or integrated into a client portal for real-time predictions.